harmless, biologically essential compound better known as table salt? Or how about hydrogen and oxygen? One is an explosive gas, and the other promotes violent combustion, yet the two combined make liquid water, which puts out fires.

Amid these chemical confabulations we find elements significant to the cosmos, allowing me to offer the Periodic Table as viewed through the lens of an astrophysicist.

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With only one proton in its nucleus, hydrogen is the lightest and simplest element, made entirely during the big bang. Out of the ninety-four naturally occurring elements, hydrogen lays claim to more than two-thirds of all the atoms in the human body, and more than ninety percent of all atoms in the cosmos, on all scales, right on down to the solar system. Hydrogen in the core of the massive planet Jupiter is under so much pressure that it behaves more like a conductive metal than a gas, creating the strongest magnetic field among the planets. The English chemist Henry Cavendish discovered hydrogen in 1766 during his experiments with H_2O (*hydro-genes* is Greek for "water-forming"), but he is best known among astrophysicists as the first to calculate Earth's mass after having measured an accurate value for the gravitational constant in Newton's famous equation for gravity.

Every second of every day, 4.5 billion tons of fast-moving hydrogen nuclei are turned into energy as they slam together to make helium within the fifteen-million-degree core of the Sun.

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Helium is widely recognized as an over-the-counter, low-density gas that, when inhaled, temporarily increases the vibrational frequency of your windpipe and larynx, making you sound like Mickey Mouse. Helium is the second simplest and second most abundant element in the universe. Although a distant second to hydrogen in abundance, there's four times more of it than all other elements in the universe combined. One of the pillars of big bang cosmology is the prediction that in every region of the cosmos, no less than about ten percent of all atoms are helium, manufactured in that percentage across the well-mixed primeval fireball that was the birth of our universe. Since the thermonuclear fusion of hydrogen within stars gives you helium, some regions of the cosmos